WASP-43b

R-0650

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-43_170306` · created 2026-07-25T22:03:41+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457818.7965 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.033 +/- 0.059
Transit depth (Rp/Rs)^2	0.11 +/- 0.39 [%]
Orbital Inclination (inc)	77.62 +/- 2.92
Airmass coefficient 1 (a1)	19.39 +/- 9.67
Airmass coefficient 2 (a2)	-2.09 +/- 0.6
Scatter in the residuals of the lightcurve fit is	50.96 %
Best Comparison Star	None
Optimal Aperture	1.22
Optimal Annulus	4.5
Transit Duration (day)	0.015 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.033 ± 0.059 vs 0.1594 (deviation 2.05σ)
Duration fit vs published	0.015 d vs 0.0512 d (deviation 2.04σ)
Mid-time O–C	-2.5 min
Depth SNR	0.5
Residual scatter	50.96 %

Light curve

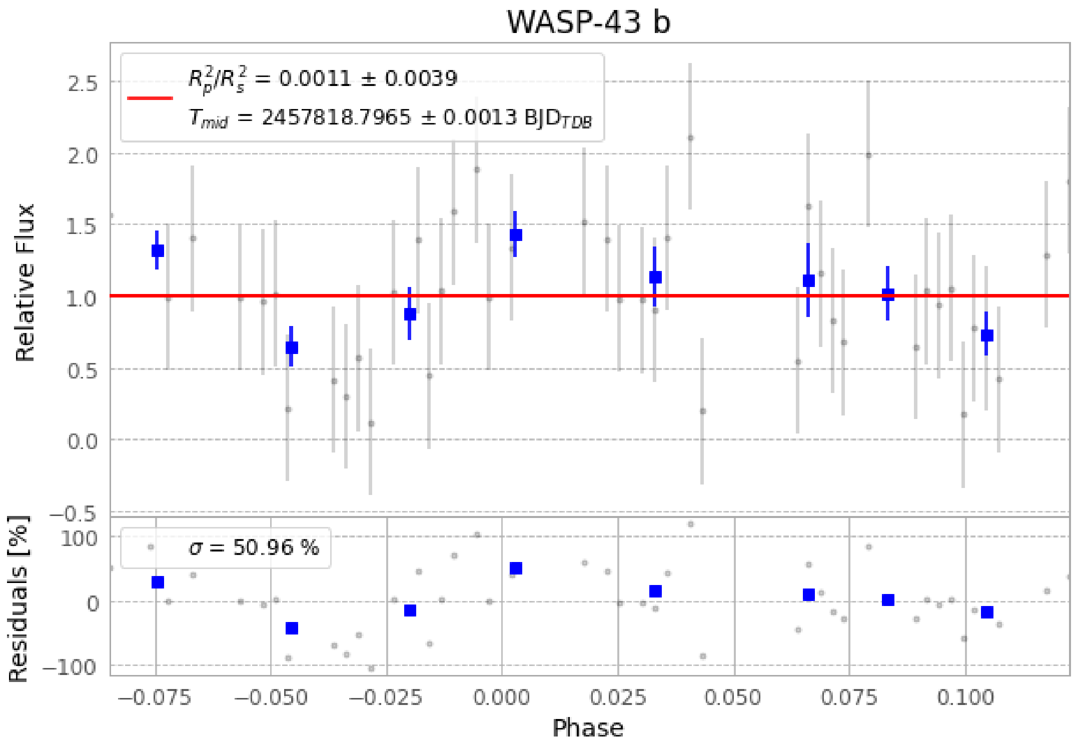

Figure 1 — FinalLightCurve_WASP-43 b_2017-03-05.png (SHA-256 5422d49b6aab4b5a…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [317, 424], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 5657d4991a009790…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494`

Data provenance

83 FITS frames (55 MB), `WASP-43170306051512.FITS` → `WASP-43170306092112.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-170306.log (a7a118976851…), FinalLightCurve_WASP-43 b_2017-03-05.png (5422d49b6aab…), FinalLightCurve_WASP-43 b_2017-03-05.csv (9893475a8cd0…)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)